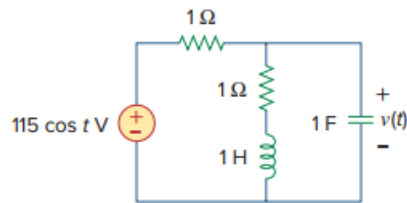


Problem

Find $v(t)$ in the RLC circuit of Fig. 9.48.



Step-by-step solution

Step 1 of 3

Refer to Figure 9.48 in the text book for the circuit.

The voltage source $v_s(t)$ is,

$$v_s(t) = 115 \cos t \text{ V}$$

The angular velocity ω is 1 rad/s.

Convert the voltage into phasor domain.

$$\mathbf{V}_s = 115 \angle 0^\circ \text{ V}$$

Determine the capacitive impedance, $\mathbf{Z}_C$.

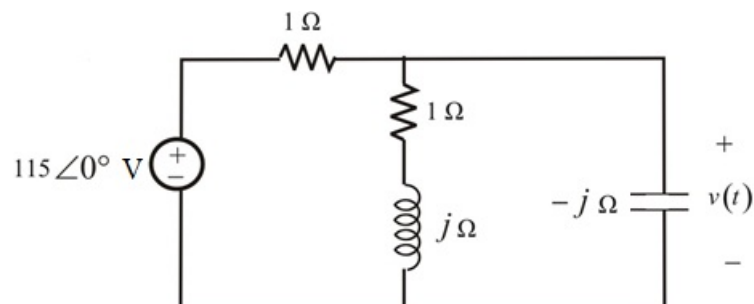
$$\begin{aligned} \mathbf{Z}_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j1 \times 1} \\ &= -j \Omega \end{aligned}$$

Determine the inductive impedance, $\mathbf{Z}_L$.

$$\begin{aligned} \mathbf{Z}_L &= j\omega L \\ &= j1 \times 1 \\ &= j \Omega \end{aligned}$$

Step 2 of 3

Draw the frequency domain equivalent circuit.



Step 3 of 3

Determine the equivalent impedance $\mathbf{Z}_{\text{eq1}}$ of the parallel combination of elements in the network.

$$\begin{aligned}\mathbf{Z}_{\text{eq1}} &= (1+j) \parallel (-j) \\ &= \frac{(1+j)(-j)}{1+j-j} \\ &= 1-j \, \Omega\end{aligned}$$

Apply voltage division rule to find the voltage, $\mathbf{v}(t)$ in phasor domain.

$$\begin{aligned}\mathbf{V} &= (115\angle 0^\circ) \left(\frac{\mathbf{Z}_{\text{eq1}}}{\mathbf{Z}_{\text{eq1}} + 1} \right) \\ &= (115\angle 0^\circ) \left(\frac{1-j}{1-j+1} \right) \\ &= 69 - j23 \, \text{V} \\ &= 72.73\angle -18.43^\circ \, \text{V}\end{aligned}$$

Convert the voltage into time domain.

$$\mathbf{v}(t) = 72.73 \cos(t - 18.43^\circ) \, \text{V}$$

Thus, the voltage $\mathbf{v}(t)$ in the circuit is $\boxed{72.73 \cos(t - 18.43^\circ) \, \text{V}}$.